THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 7: **Particulate Nature of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The melting point of ice is: 1

(a) 0 degrees Celsius **(b)** 37 degrees Celsius
(c) 100 degrees Celsius **(d)** -10 degrees Celsius

2. We sweat on a hot day. Sweat evaporates and: 1

(a) makes it heavy **(b)** makes it dry **(c)** cools the body **(d)** heats the body

3. Diffusion is fastest in: 1

(a) all equally **(b)** solids **(c)** liquids **(d)** gases

4. The boiling point of water at normal atmospheric pressure is: 1

(a) 212 degrees Celsius **(b)** 100 degrees Celsius
(c) 0 degrees Celsius **(d)** 50 degrees Celsius

5. Humidity is the: 1

(a) amount of oxygen **(b)** amount of water vapour in the air
(c) amount of dust in the air **(d)** amount of nitrogen

6. Particles are closest together and in fixed positions in: 1

(a) all three **(b)** gases **(c)** solids **(d)** liquids

7. The change of a gas to a liquid on cooling is called: 1

(a) melting **(b)** sublimation **(c)** condensation **(d)** evaporation

8. When humidity is high, the rate of evaporation: 1

(a) increases **(b)** is unaffected **(c)** becomes infinite **(d)** decreases

9. Which state of matter has a definite volume but no definite shape? 1

(a) gas **(b)** plasma **(c)** solid **(d)** liquid

10. Conversion of a solid directly to a gas on heating is: 1

(a) melting **(b)** condensation **(c)** evaporation **(d)** sublimation

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do we feel cool when we put acetone or perfume on our hand? 2

12. Why does a drop of ink spread faster in hot water? 2

13. Why does a gas fill its container completely? 2

14. What is sublimation? Give an example. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Compare the arrangement and motion of particles in solids, liquids and gases. 3

16. Why does ice at 0 degrees Celsius cool a drink more than water at 0 degrees Celsius? 3

17. List three examples of sublimation and three of condensation. 3

OR

State three properties of solids, liquids and gases (one each). 3

18. Why does steam at 100 degrees Celsius cause more severe burns than boiling water at 100 degrees Celsius? 3

OR

Why does the smell of hot food reach us faster than that of cold food? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the changes of state with names and examples. 5

OR

Explain the properties of the three states of matter in terms of the forces between particles and how they explain the changes of state. 5

20. Explain evaporation and the factors affecting it. State three daily-life applications. 5

OR

Explain how humidity, wind and temperature influence drying of clothes and design a fair test. 5

END OF QUESTION PAPER — Total: 40 Marks

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